Band-1 Hadamard Order 5284 — Miyamoto (Double Paley-I + Paley-I-direct seeds H_1320); Miyamoto-only bonus NOT in SPK; Gap 2052 Still OPEN

H_660/H_1320/C_1322/H_5284 PASS; Paley-I-direct H_1320 PASS; Paley-II H_5284 SKIP; 5284 Miyamoto-only NOT in SPK; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_5188_Miyamoto`; `2026-09-15_Band1_Hadamard_5156_Miyamoto` (Miyamoto-only pattern).
Lab. `/workspace/experiments/band1/hadamard_5284_miyamoto/`.
Cutoff. 2026-09-15T19:27:22Z (~3:27PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 660, 1320 and 5284 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double(Paley-I) seed H_{1320} (H_{660} = Paley-I($q = 659$); H_{1320} = Sylvester-double(H_{660})), with alternate Paley-I direct ($q = 1319$). Paley-II path for H_{1320} is **FAIL** ($q = 659 \equiv 3 \pmod 4$; needs $\equiv 1$). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 5284 is a **Miyamoto-only bonus NOT in SPK**: Paley-I FAIL ($q = 5283 = 3^2 \cdot 587$), Paley-II FAIL ($q = 2641 = 19 \cdot 139$), Kronecker FAIL (4×1321). Same honesty pattern as 5156/4916/4868 (Miyamoto bonus). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 5188 on box. This increment ships **order 5284** ($5284 = 4 \cdot 1321$, 1321 prime $\equiv 1 \pmod 4$) via Miyamoto with a **Double(Paley-I) seed** H_{1320} and verified **Paley-I direct** alternate. Symmetric Paley conference C_{1322} over prime 1321; $H_{5284} = \text{Miyamoto}(H_{1320}, C_{1322})$. Classical Paley-II path for order 5284 ($q = 2641$) is **SKIP** (composite). Unit tests **69/69 PASS**. SPK next gap remains **2052** (OPEN). Order 5284 is a **Miyamoto-only bonus NOT in SPK**. Seeds H_{660}/H_{1320} enter SPK via Double Paley-I / Paley-I direct. Prior 5188 remains in SPK via Paley-II. The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 5284; Miyamoto; Paley type I; Double Paley-I; Paley conference; constructive gap 2052; SPK; Band-1; Miyamoto-only bonus; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{660} constructed and verified (Paley type I, $q = 659$). 2. H_{1320} constructed and verified (Sylvester double of H_{660}). Paley-II path for H_{1320} **FAIL** ($q = 659 \equiv 3 \pmod 4$). 3. H_{1320} also constructed via Paley-I direct ($q = 1319$) and verified (alternate). 4. C_{1322} constructed and verified (Paley conference, $q = 1321$): $CC^\top = 1321I$, zero diagonal. 5. H_{5284} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 6. SPK: $5284 \notin$ SPK (Miyamoto-only). Paley-I / Paley-II / Kronecker paths for 5284 **rejected**. Next constructive gap remains **2052** (OPEN). Verified H_{660}/H_{1320} enter SPK as Double Paley-I / Paley-I-direct seeds. Prior 5188 remains in SPK via Paley-II ($q = 2593$). Prior 5156 remains Miyamoto-only (excluded).

2. Arithmetic and eligibility

Quantity	Value
Target order	$5284 = 4 \cdot 1321$
$q = 1321$	prime, $1321 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{1320}
Needed conference	order $1322 = q + 1$
H_{1320} path (used)	Double(Paley-I): $H_{660} =$ Paley-I($q = 659$); $H_{1320} =$ Sylvester-double(H_{660})
H_{1320} Paley-I direct	**PASS**: $q = 1319$ prime $\equiv 3 \pmod 4$ (alternate/verify)
H_{1320} Paley-II	**FAIL**: $2(q + 1) = 1320 \Rightarrow q = 659 \equiv 3 \pmod 4$ (needs $\equiv 1$)

Quantity	Value
Conference path	Symmetric Paley over prime 1321
Paley-I for 5284	**REJECTED**: $q' = 5283 = 3^2 \cdot 587$ not prime
Paley-II for 5284	**REJECTED**: $q' = 2641 = 19 \cdot 139$ not prime
Kronecker for 5284	**REJECTED**: only 4×1321; $1321 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{660}	Paley type I, $q = 659$	0	0.3454	**PASS**
H_{1320}	Sylvester double of H_{660}	0	1.0487	**PASS**
H_{1320} Paley-I direct	Paley type I, $q = 1319$	0	1.8933	**PASS**
H_{1320} Paley-II	$q = 659 \equiv 3 \pmod 4$	—	—	**FAIL** (needs $\equiv 1$)
C_{1322}	Paley conference, $q = 1321$	0	1.8676	**PASS**
H_{5284}	Miyamoto(H_{1320}, C_{1322})	0	71.292	**PASS**
H_{5284} Paley-II	$q = 2641 = 19 \cdot 139$	—	—	**SKIP** (not prime)

Primary pipeline wall (incl. unit tests + SPK refresh) $\approx$ 78.3028 s. Compressed ± 1 matrices saved as `H660_pm1.npz`, `H1320_pm1.npz`, `H1320_paleyI_direct_pm1.npz`, `C1322_pm1.npz`, `H5284_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior classical/SPK ships (through 5188/5156/4948/4804/… + **648**/**1296** Double Paley-I + **5188** Paley-II; **exclude** Miyamoto-only 5284/5156/4916/4868/4612/4516/4468/438 $\cup$ verified H_{660}/H_{1320} (Double Paley-I / Paley-I direct) $\cup$ Sylvester $\cup$ Paley I/II (incl. prime powers), closed under Kronecker (limit 5600). Order 5284 is **NOT** folded into SPK (Miyamoto-only).

Quantity	Value
Known count	1140
5284 in SPK	**False** (Miyamoto-only)
5188 in SPK	**True** (prior Paley-II q=2593)
1320 in SPK	**True** (Double Paley-I / Paley-I direct)
660 in SPK	**True** (Paley-I seed)
1296/648 in SPK	**True** (prior 5188 seeds)
Next constructive gap	**2052** (OPEN)

5. Gap 2052 (method triage / non-finding ONLY)

Paley-I, Paley-II, Kronecker, and Miyamoto-with-Paley-conference paths for order 2052 all **FAIL** on-box arithmetic. This is a **non-finding**, not a counterexample. Gap **2052 remains OPEN**. Conjecture **OPEN**.

6. Non-claims

- No full Hadamard conjecture claim.
- `clay_closed = false` forever in this ship.
- GATE-04 collection **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- Gap 2052 **not** filled.
- Order 5284 is a **finite Miyamoto-only bonus**, not an SPK classical path.

7. Reproduce

```
cd /workspace/experiments/band1/hadamard_5284_miyamoto
python3 verify_hadamard_5284.py
```

Cutoff: 2026-09-15T19:27:22Z (~3:27PM ET).